Public Libraries in Maine: A Geospatial Research Project

CPT 127 - Exploring Geospatial Data with GeoPandas
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Accompanying code and data files located at:
https://github.com/sschlotterbeck/cpt_127_assignments/tree/Libraries-in-Maine-Project

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"There is not such a cradle of democracy upon the earth as the Free Public Library, this republic of letters, where neither rank, office, nor wealth receives the slightest consideration."

--Andrew Carnegie

Introduction

In addition to a deep, personal love of public libraries, I believe that they are also one of our most important democratizing civic institutions, and so I decided to research the distribution of libraries for this project. My research question was:

> a) how does the distribution of public libraries in Maine compare to the distribution of people in in the state?
> b) is there an equitable distribution of libraries in regards to people living in more urban areas compared to more rural areas?

The data I used in this project came from:
1) The Maine GeoLibrary website[i]
2) The Maine State Library website[ii]
3) Wikipedia (which cites the various sources for the data it displays on its pages, including the U.S. Census Bureau)[iii]

Approach and Analysis

I found a database of libraries in Maine on the Maine GeoLibrary website, and after being unsuccessful in my attempts to successfully format a request string or locate a GeoJSON link, I discovered that I could utilize the query form on the website to generate a GeoJSON-formatted page, which I then downloaded, and started my analysis by reading that saved file into GeoPandas.

Upon examining that database, I noticed that many of the libraries listed weren't public libraries—a number were either law libraries, private libraries at universities, or medical libraries at hospitals, so I searched for a list of public libraries in Maine to filter this initial dataset, which I found at the Maine State Library website.

I pasted the table of public libraries and the county they are in from that site into a spreadsheet and saved it as a csv file so I could then load it into Pandas and GeoPandas.

I cursorily cleaned up the data from both datasets, and then did an initial check for overlap of libraries in both datasets. Out of the total 261 public libraries listed on the Maine State Library website, a check showed a total of 218 matches in the Maine GeoLibrary library database.

I proceeded to examine the remaining unmatched libraries for potential overlap due to different spelling, punctuation, etc. After spotting a few consistent patterns that were easily cleared up (ex. the Maine State Library dataset placed periods after initials in names, while the Maine GeoLibrary dataset did not), I ended up cross-referencing each remaining library with a Google search to double-check the correct spelling, alternate names for the same library on the other list, etc. Once I made it through all of the non-matching libraries that looked to be likely candidates for being public libraries, I only had 3 left from the Maine State Library list of public libraries that I couldn't find in the Maine GeoLibrary dataset. Two of those libraries only opened within the past few years, so it seems possible they weren't open at the time the dataset was created.

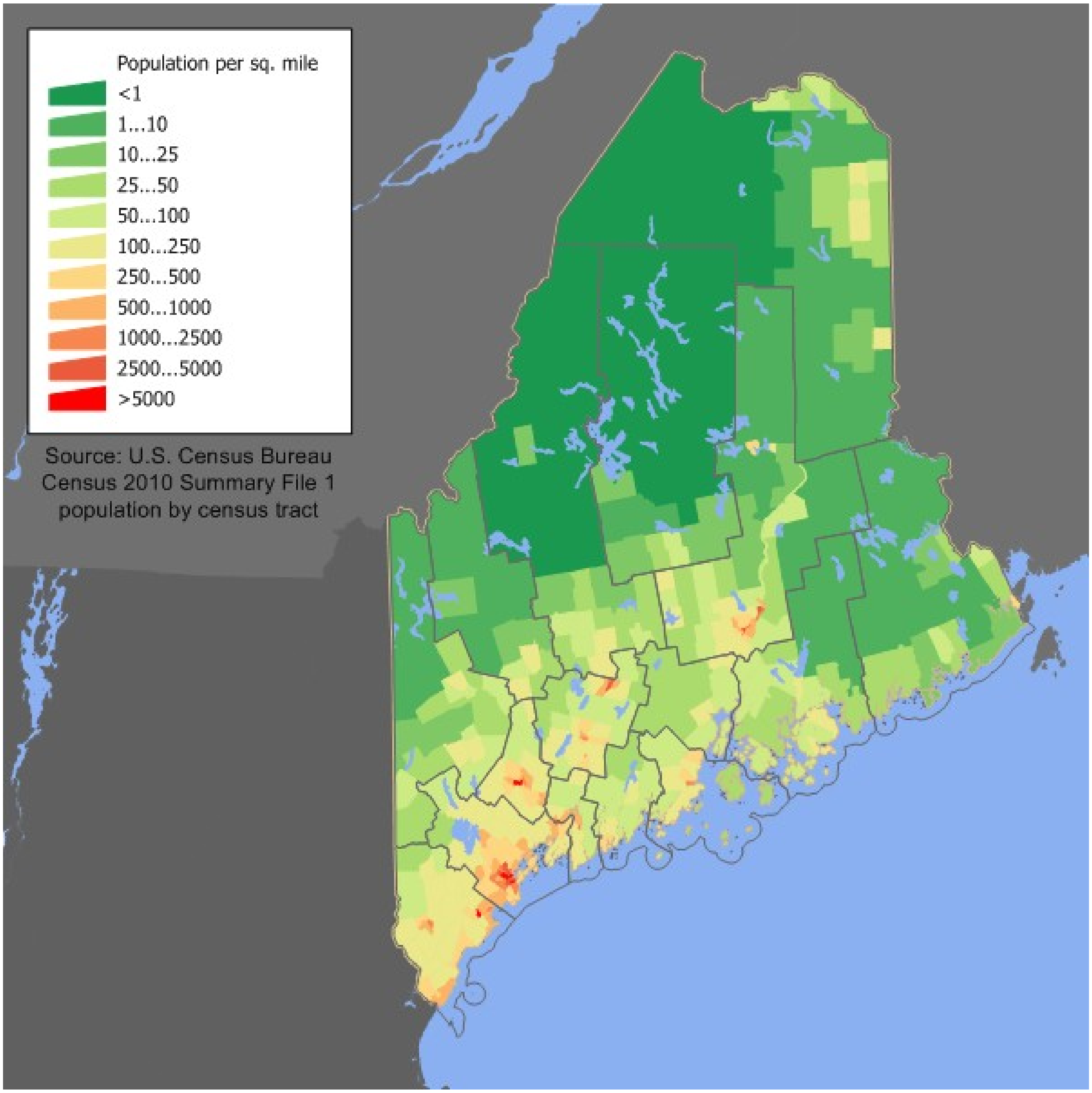
At this point, I updated the code in my python file to harmonize the library names in both datasets so that when checking them against each other, all the correct libraries would be identified, and then plotted that dataset. I initially attempted to plot the library onto a state boundary outline of Maine that I also found in the Maine GeoLibrary[iv] to more easily overlay it over a population density map of Maine I found on Wikipedia[v], but the dataset appeared to be incomplete when plotted, and I eventually gave up on that and eyeballed it as best I could lining up the library scatter plot onto the population density map.

To generate bar graphs, I made a pandas DataFrame that contained various data broken down by county, and then plotted that using the .plot() method with Matplotlib in python.

Findings

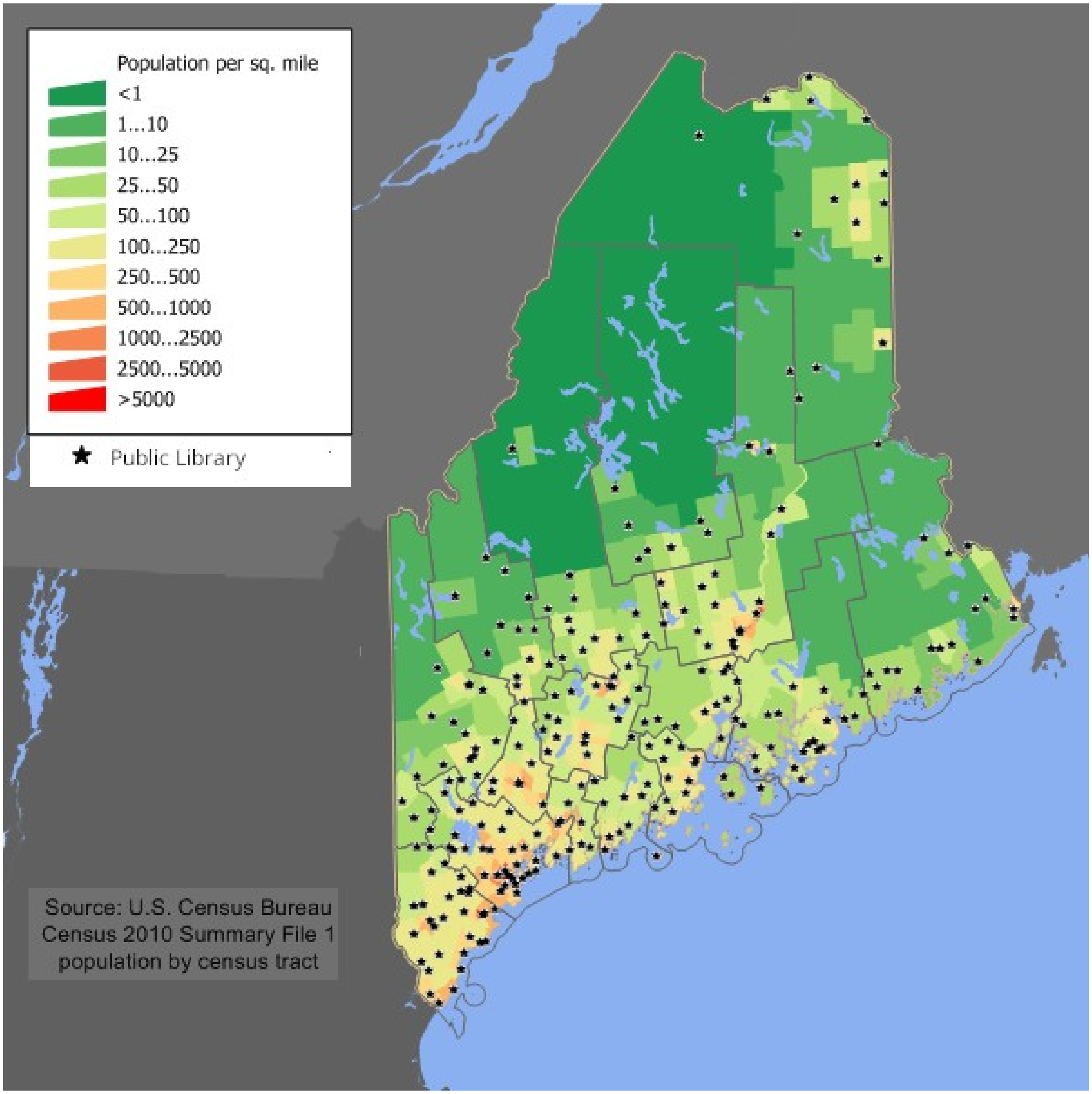
As you can see on the maps on the following two pages, on a statewide level, Maine's public libraries pretty closely follow the areas with the highest population density:

Fig. 1



Maine State Population density map (source: Wikipedia[vi])

Fig. 2



Distribution of Public Libraries in Maine (overlaid on Population Density Map)[vii]

In terms of further analysis broken down by county: though it likely comes as no surprise that the counties with higher population densities also generally have the highest library densities per square mile, with some exceptions (see Fig. 3), when examining the Library to Population Density Ratios of the various counties (ie the average number of libraries per person in any given square mile of a county), less densely populated counties actually make a pretty strong showing across the state (Fig. 4).[viii]

Fig. 3

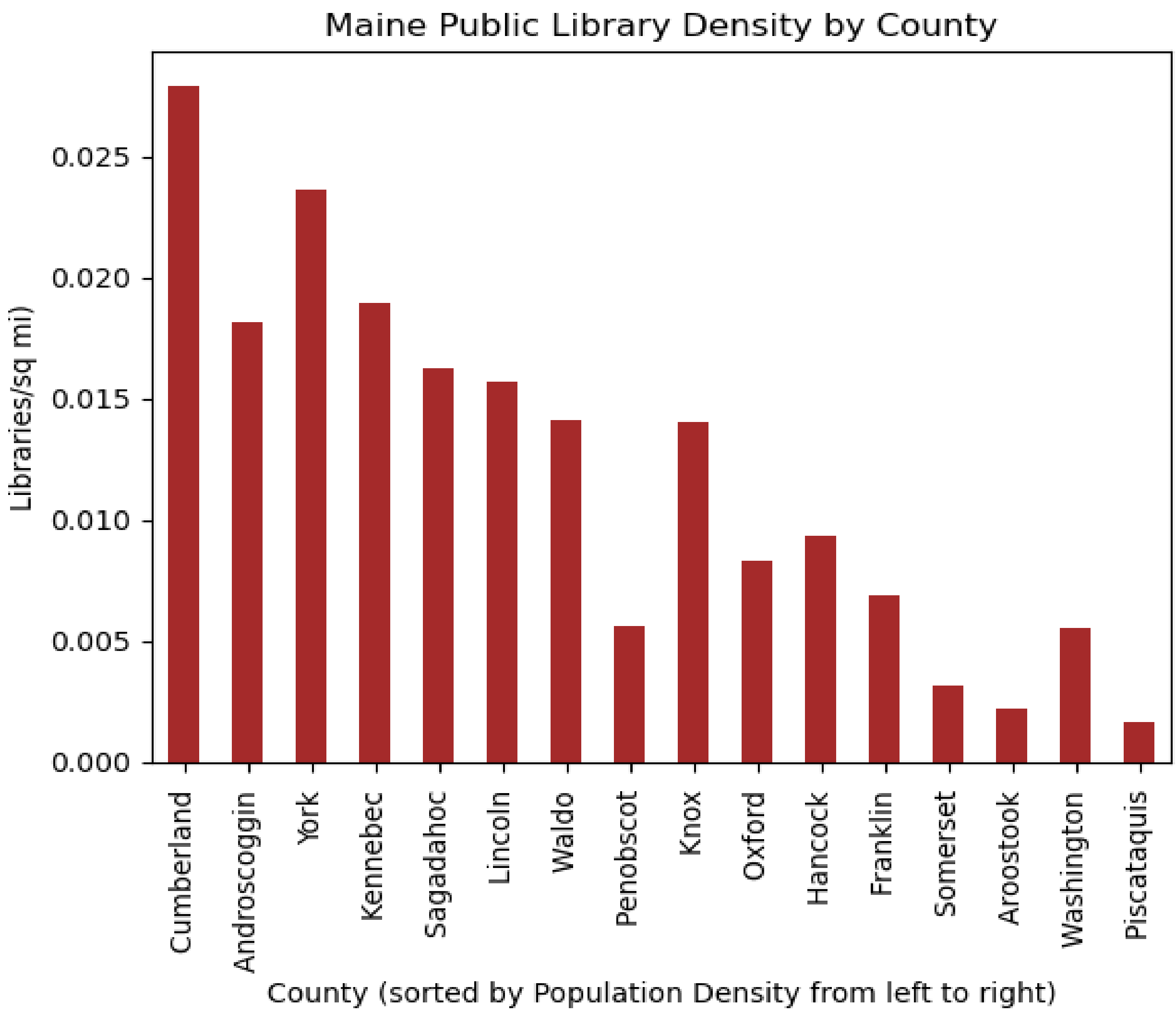
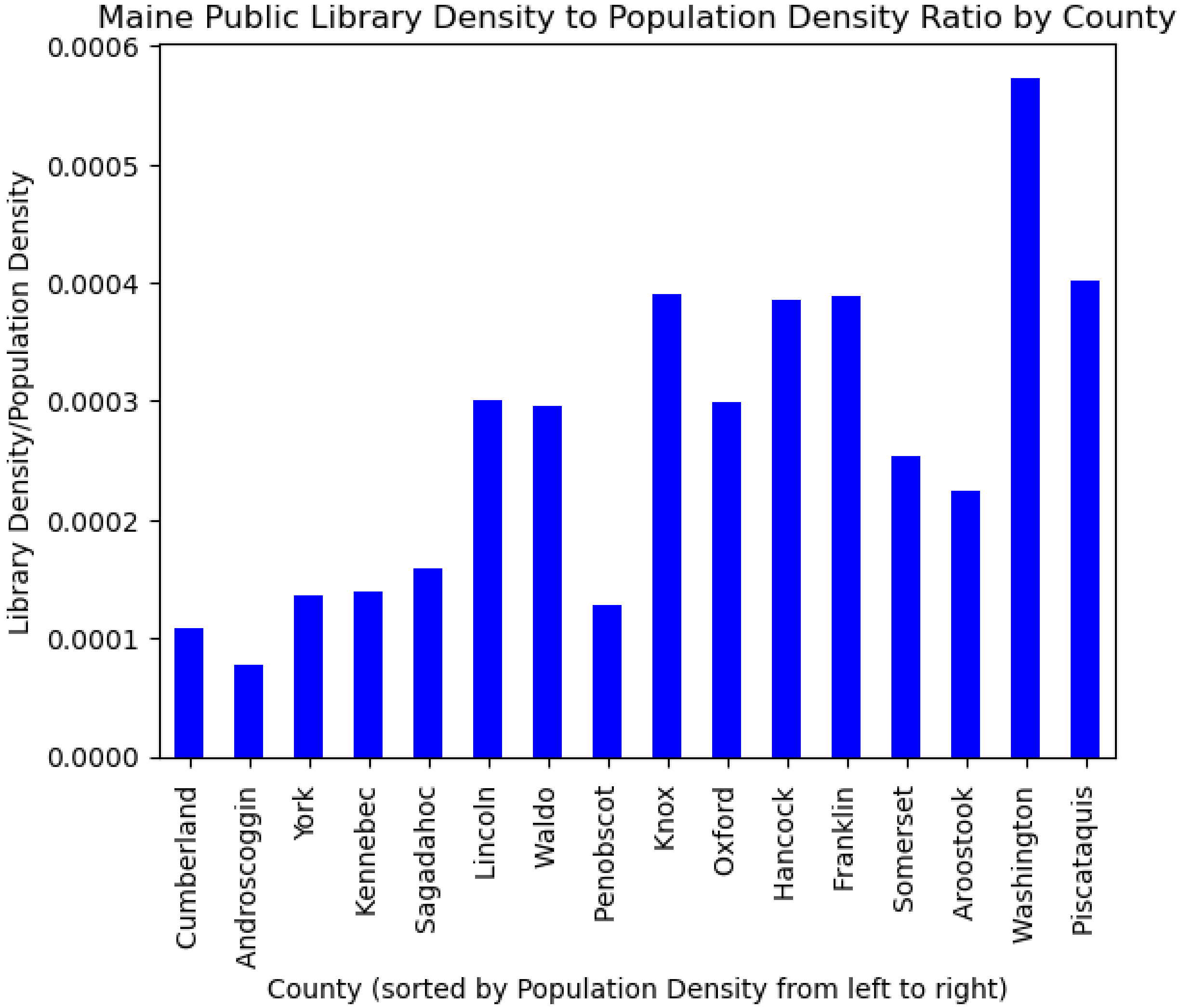


Fig. 4



There are a few anomalies in the Library Density graph (Androscoggin and Penobscot counties for example), and I would be curious about what the data might show if one followed up by bringing in per capita income or other economic metrics to analyze alongside the current data (for example, Cumberland and York counties are first and third, respectively, for highest per capita incomes in the state, while Androscoggin is ninth, perhaps that is an explanation for the disparity in library density)[ix].

In summary, the locations of Maine's public libraries pretty closely follow the distribution of people across the state. If you live in one of the counties in Maine with higher population densities, your nearest public library will be, on average, located physically closer to you than if you live in one of the counties with lower population densities, but even in less densely populated counties, the humans and libraries tend to cluster together.

In terms of how many other people in your area you are sharing your nearest public library with, the number fluctuates more widely, but the lowest number of people sharing a single library is in fact in a less densely populated county.

i "Maine Libraries GeoLibrary", https://maine.maps.arcgis.com/home/item.html?id=de89492ad04d49fe8e5761d0c7b7b1da –accessed July 17, 2025

ii "Maine State Library – Public Library Directory", https://www.maine.gov/msl/libs/directories/public.shtml –accessed July 17, 2025

iii "List of counties in Maine", https://en.wikipedia.org/wiki/List_of_counties_in_Maine –accessed July 17, 2025

iv https://services1.arcgis.com/RbMX0mRVOFNTdLzd/ArcGIS/rest/services/Maine_State_Boundary_Line/FeatureServer

v https://upload.wikimedia.org/wikipedia/commons/3/3d/Maine_population_map.png

vi https://upload.wikimedia.org/wikipedia/commons/3/3d/Maine_population_map.png

vii Code for Maine public library distribution plot located at: https://github.com/sschlotterbeck/cpt_127_assignments/blob/Libraries-in-Maine-Project/public_libraries_in_maine_file_1.py

viii Code for bar graphs located at: https://github.com/sschlotterbeck/cpt_127_assignments/blob/Libraries-in-Maine-Project/public_libraries_in_maine_file_2.py

ix https://en.wikipedia.org/wiki/List_of_Maine_locations_by_per_capita_income